DNS & DHCP Server Lab

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1. Project Overview

This lab deploys a fully functional DNS and DHCP server on Ubuntu Server 24.04 LTS running inside VirtualBox on macOS. BIND9 provides authoritative DNS resolution for the lab.local domain, while ISC-DHCP-Server manages dynamic IP allocation for client devices. The server is accessed and managed remotely via SSH — simulating real-world Linux server administration.

Lab Environment

Host Machine	macOS (Apple Silicon ARM64)
Hypervisor	Oracle VirtualBox 7.1+
Server OS	Ubuntu Server 24.04.4 LTS (aarch64)
DNS Software	BIND9 (named) v9.18.39
DHCP Software	ISC-DHCP-Server v4.4.3-P1
Network	VirtualBox Host-only (192.168.56.0/24)
Server IP	192.168.56.2 (static via Netplan)
DHCP Range	192.168.56.100 – 192.168.56.200
DNS Zone	lab.local
Remote Access	SSH (OpenSSH) from Mac Terminal

2. Network Design

Host	IP Address	Role	Interface
DNS-DHCP-SER	192.168.56.2	DNS + DHCP Server	enp0s9 (Host-only)
DNS-DHCP-SER	10.0.2.15	NAT (internet)	enp0s8 (NAT)
Client VMs	192.168.56.100+	DHCP Clients	Host-only
VirtualBox Host	192.168.56.1	Default Gateway	vboxnet0

3. DNS Configuration (BIND9)

BIND9 is configured as an authoritative DNS server for the lab.local domain. The named.conf.options file enables recursion and forwards external queries to Google DNS (8.8.8.8). A forward zone file defines A records for all lab hosts.

named.conf.options

```
options {
directory "/var/cache/bind";
recursion yes;
allow-query { any; };
forwarders { 8.8.8.8; 8.8.4.4; };
dnssec-validation no;
listen-on { any; };
};
```

Zone file: /etc/bind/zones/db.lab.local

```
$TTL 604800
@ IN SOA ns1.lab.local. admin.lab.local. ( 2024010101 )
@ IN NS ns1.lab.local.
ns1 IN A 192.168.56.2
server IN A 192.168.56.2
client IN A 192.168.56.100
```

4. DHCP Configuration (ISC-DHCP)

ISC-DHCP-Server manages dynamic IP allocation for all clients on the 192.168.56.0/24 subnet. Each lease includes DNS server, domain name, default gateway, and broadcast address — matching a real enterprise DHCP deployment.

/etc/dhcp/dhcpd.conf

```
default-lease-time 86400;
max-lease-time 172800;
authoritative;

subnet 192.168.56.0 netmask 255.255.255.0 {
range 192.168.56.100 192.168.56.200;
option domain-name-servers 192.168.56.2;
option domain-name "lab.local";
option routers 192.168.56.1;
option broadcast-address 192.168.56.255;
}
```

5. Verification — BIND9 Status

BIND9 is active and running. The logs confirm it is listening on 192.168.56.2#53 (host-only interface) and 10.0.2.15#53 (NAT interface) — ready to serve DNS queries from all lab clients.

```
admin@DNS-DHCP-SER:~$ sudo systemctl status bind9
● named.service - BIND Domain Name Server
     Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-03-19 19:59:37 UTC; 48min ago
       Docs: man:named(8)
   Main PID: 634 (named)
     Status: "running"
      Tasks: 4 (limit: 2207)
     Memory: 31.9M (peak: 32.2M)
        CPU: 36ms
     CGroup: /system.slice/named.service
             └─634 /usr/sbin/named -f -u bind

Mar 19 20:01:55 DNS-DHCP-SER named[634]: no longer listening on 10.0.2.15#53
Mar 19 20:01:55 DNS-DHCP-SER named[634]: listening on IPv4 interface enp0s8, 10.0.2.15#53
Mar 19 20:01:56 DNS-DHCP-SER named[634]: listening on IPv6 interface enp0s8, fd17:625c:f037:2:a00:27ff:feb1:2793#53
Mar 19 20:20:07 DNS-DHCP-SER named[634]: listening on IPv6 interface enp0s9, fe80::a00:27ff:fe47:ec28%3#53
Mar 19 20:20:07 DNS-DHCP-SER named[634]: no longer listening on 10.0.2.15#53
Mar 19 20:20:07 DNS-DHCP-SER named[634]: no longer listening on fd17:625c:f037:2:a00:27ff:feb1:2793#53
Mar 19 20:20:07 DNS-DHCP-SER named[634]: listening on IPv4 interface enp0s9, 192.168.56.2#53
Mar 19 20:20:07 DNS-DHCP-SER named[634]: listening on IPv4 interface enp0s8, 10.0.2.15#53
Mar 19 20:20:07 DNS-DHCP-SER named[634]: listening on IPv6 interface enp0s8, fd17:625c:f037:2:a00:27ff:feb1:2793#53
Mar 19 20:20:09 DNS-DHCP-SER named[634]: listening on IPv6 interface enp0s9, fd87:a03b:17b8:73ac:a00:27ff:fe47:ec28#53
admin@DNS-DHCP-SER:~$
```

Figure 1: systemctl status bind9 — active (running), listening on 192.168.56.2#53

6. Verification — ISC-DHCP Status

ISC-DHCP-Server is active and running on interface enp0s9 (192.168.56.0/24). The server is listening for DHCP discover broadcasts and ready to assign leases from the configured pool.

```
admin@DNS-DHCP-SER:~$ sudo systemctl status isc-dhcp-server
● isc-dhcp-server.service - ISC DHCP IPv4 server
     Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-03-19 20:46:39 UTC; 1min 41s ago
       Docs: man:dhcpd(8)
   Main PID: 2580 (dhcpd)
      Tasks: 1 (limit: 2207)
     Memory: 3.7M (peak: 4.0M)
        CPU: 6ms
     CGroup: /system.slice/isc-dhcp-server.service
             └─2580 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /etc/dhcp/dhcpd.conf enp0s9

Mar 19 20:46:39 DNS-DHCP-SER dhcpd[2580]: Wrote 0 leases to leases file.
Mar 19 20:46:39 DNS-DHCP-SER sh[2580]: PID file: /run/dhcp-server/dhcpd.pid
Mar 19 20:46:39 DNS-DHCP-SER sh[2580]: Wrote 0 leases to leases file.
Mar 19 20:46:39 DNS-DHCP-SER dhcpd[2580]: Listening on LPF/enp0s9/08:00:27:47:ec:28/192.168.56.0/24
Mar 19 20:46:39 DNS-DHCP-SER sh[2580]: Listening on LPF/enp0s9/08:00:27:47:ec:28/192.168.56.0/24
Mar 19 20:46:39 DNS-DHCP-SER sh[2580]: Sending on   LPF/enp0s9/08:00:27:47:ec:28/192.168.56.0/24
Mar 19 20:46:39 DNS-DHCP-SER sh[2580]: Sending on   Socket/fallback/fallback-net
Mar 19 20:46:39 DNS-DHCP-SER dhcpd[2580]: Sending on   LPF/enp0s9/08:00:27:47:ec:28/192.168.56.0/24
Mar 19 20:46:39 DNS-DHCP-SER dhcpd[2580]: Sending on   Socket/fallback/fallback-net
Mar 19 20:46:39 DNS-DHCP-SER dhcpd[2580]: Server starting service.
admin@DNS-DHCP-SER:~$
```

Figure 2: systemctl status isc-dhcp-server — active (running) on enp0s9/192.168.56.0/24

7. Verification — DNS Resolution

The dig command confirms successful DNS resolution. The ANSWER SECTION returns 192.168.56.2 for server.lab.local with a query time of 0ms — confirming the authoritative zone is loaded and responding correctly.

```
admin@DNS-DHCP-SER:~$ dig server.lab.local @192.168.56.2

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> server.lab.local @192.168.56.2
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 53135
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 01976f75ebb3271d0100000069bc612d6f9853f72dfce38e (good)
;; QUESTION SECTION:
;server.lab.local.              IN      A

;; ANSWER SECTION:
server.lab.local.       604800  IN      A       192.168.56.2

;; Query time: 0 msec
;; SERVER: 192.168.56.2#53(192.168.56.2) (UDP)
;; WHEN: Thu Mar 19 20:48:45 UTC 2026
;; MSG SIZE  rcvd: 89

admin@DNS-DHCP-SER:~$
```

Figure 3: dig server.lab.local @192.168.56.2 — ANSWER: 192.168.56.2 (0ms query time)

8. Verification — DHCP Config File

The dhcpd.conf file shows the clean, validated configuration with correct subnet declaration, IP range, DNS options, domain name, and gateway — all properly formatted and tested with dhcpd -t before deployment.

```
admin@DNS-DHCP-SER:~$ cat /etc/dhcp/dhcpd.conf
default-lease-time 86400;
max-lease-time 172800;
authoritative;

subnet 192.168.56.0 netmask 255.255.255.0 {
    range 192.168.56.100 192.168.56.200;
    option domain-name-servers 192.168.56.2;
    option domain-name "lab.local";
    option routers 192.168.56.1;
    option broadcast-address 192.168.56.255;
}
admin@DNS-DHCP-SER:~$
```

Figure 4: cat /etc/dhcp/dhcpd.conf — validated configuration file

9. Verification — DHCP Leases File

The dhcpd.leases file is initialised and active. The server DUID has been generated confirming the DHCP daemon is operational. Leases will populate as client VMs connect and request IP addresses from the pool.

```
admin@DNS-DHCP-SER:~$ cat /var/lib/dhcp/dhcpd.leases
# The format of this file is documented in the dhcpd.leases(5) manual page.
# This lease file was written by isc-dhcp-4.4.3-P1

# authoring-byte-order entry is generated, DO NOT DELETE
authoring-byte-order little-endian;

server-duid "\000\001\000\00110\035/\010\000'G\354(";
```

Figure 5: cat /var/lib/dhcp/dhcpd.leases — server DUID generated, ready for leases

10. Troubleshooting Log

Real-world issues encountered and resolved during this lab — demonstrating diagnostic skills beyond basic configuration:

VirtualBox Boot Loop	VM booted to blank boot menu. Fixed by attaching Ubuntu ARM64 ISO to optical drive and setting boot order to Optical first.
Wrong ISO Architecture	AMD64 ISO failed on Apple M-series Mac. Resolved by downloading Ubuntu Server ARM64 ISO specifically for aarch64.
Host-only Adapter Missing	Network adapter dropdown empty. Fixed by creating vboxnet0 manually in VirtualBox Network Manager.
Interface enp0s9 DOWN	Host-only adapter had no IP and was in DOWN state. Resolved by assigning static IP via Netplan (50-cloud-init.yaml) and running netplan apply.
DHCP Config Corruption	Lease time values triggered 'exceeds max (255)' error — caused by invisible Unicode characters from copy-paste. Fixed by writing config using echo pipe commands to avoid character encoding issues.
SSH Connection Refused	OpenSSH not installed by default. Resolved by installing openssh-server and enabling the service — enabling remote management from Mac Terminal.
DNSSEC Validation Failure	External DNS forwarding failed due to DNSSEC root server unreachability in NAT-isolated VM. Resolved by setting dnssec-validation no in named.conf.options.

11. Skills Demonstrated

Linux Admin	`Ubuntu Server installation, user management, service control via systemctl`
DNS (BIND9)	`Authoritative zone config, forward records, named-checkzone validation`
DHCP	`ISC-DHCP scope config, lease management, DHCP options (DNS, gateway, domain)`
Networking	`Static IP via Netplan, interface management, host-only networking`
SSH	`Remote server management via OpenSSH from Mac Terminal`
Troubleshooting	`7 real issues diagnosed and resolved — architecture, encoding, networking`
Documentation	`Config files, verification screenshots, troubleshooting log`